

P2P CBDC Offline Wallet Platform

Domain: Retail Payments / Central Bank Digital Currency | Statutory Architecture

Parameter	Evaluation Result	Governance & Statutory Impact
MoSCoW Baseline	Should Have	Critical operational and statutory compliance requirement for Retail Payments / Central Bank Digital Currency.
Risk Classification	CRITICAL (98/100)	Mission-critical financial tier. Direct automated capital risk or physical cash exposure requiring zero-trust stage gates.
Total Requirements	35 Modular Requirements (Min 30)	Decomposed across Frontend, Backend, Auth, Security, Database Residency, and Resilience

1. System Scope & Stakeholder Requirement Statement

"RBI Regulatory Sandbox pilot cohort for offline peer-to-peer digital rupee (CBDC) payments with synthetic money, capped at 1000 users and maximum wallet balance of 2000 INR."

2. Modular Software Requirements Specification (35 Detailed Points)

Pillar: Frontend & Mobile UI/UX (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-FE-01	P2P CBDC Offline Wallet Platform — Responsive Client Interface & Intent Layer Provide intuitive, accessible digital interfaces for stakeholders of P2P CBDC Offline Wallet Platform with real-time feedback and sub-250ms screen transition latency.	Must Have	Regulation: RBI Digital Payment Security Controls Sec 4 Ref: Fintech Regulatory Sandbox Pilot Problem Statement Acceptance: Renders complete workflow with client-side form validation and latency < 250ms.
REQ-FE-02	Dynamic Multilingual Localization (12 Indian Languages) Implement dynamic multilingual localization (English, Hindi, and regional languages) with offline draft persistence and sync-on-reconnect.	Should Have	Regulation: Digital Personal Data Protection Act, 2023 (Multilingual Consent) Ref: Sebi Mutual Funds Regulations 2026 Acceptance: Allows offline caching in encrypted storage; synchronizes queued records within 2s of reconnection.
REQ-FE-03	Digital KYC Document Uploader & Real-Time Verification Deliver secure client-side document capture with image compression, edge blur detection, and instant verification feedback.	Must Have	Regulation: RBI KYC Master Direction & Information Technology Act 2000 Ref: 1556052828Rbi-Handbookofregulationsataglance-27.2.2025-3-76 Acceptance: Validates document MIME-types and enforces 5MB limit; rejects blurry uploads client-side.
REQ-FE-04	Interactive Audit Trail & Activity Dashboard Provide interactive dashboard displaying real-time transaction statuses, pending approvals, and downloadable account statements.	Should Have	Regulation: RBI Fair Practices Code & Customer Protection Norms Acceptance: Displays real-time transaction timeline with instant PDF statement download.
REQ-FE-05	Session Security & Device Fingerprint UI Controls Enforce device fingerprint binding, automated background biometric re-authentication prompts, and visual privacy watermarks on all sensitive screens.	Must Have	Regulation: RBI Digital Payment Security Controls Sec 4.3 Acceptance: Detects root/emulator status at launch; triggers graceful lock with user guidance.

Pillar: Backend Architecture & Core APIs (6 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-BE-01	Peer-to-Peer Transaction Gateway RESTful API gateway for P2P transactions between verified farmers, implementing the CBDC protocol for secure, tamper-proof transfer of digital rupees	Should Have	Regulation: RBI Guidelines on Digital Cash Instruments (DCI) - Operational Framework for Issuers Acceptance: API gateway processes 1000 successful peer-to-peer transactions over the first month, each transaction confirmed via blockchain smart contract execution
REQ-BE-02	Core Microservices & High-Throughput Switch Engine Deploy modular, low-latency microservices orchestrating primary business workflows with transactional atomicity, idempotent request handling, and REST/gRPC API contracts.	Must Have	Regulation: RBI Core Banking Architecture Guidelines Ref: Fintech Regulatory Sandbox Pilot Problem Statement Acceptance: Maintains idempotent processing with unique UUID key checks; enforces 99.95% API uptime.
REQ-BE-03	High-Throughput Asynchronous Event Streaming Pipeline Integrate an event-driven streaming pipeline (Kafka/RabbitMQ) for asynchronous notification dispatch, ledger state transitions, and audit telemetry.	Must Have	Regulation: RBI Cyber Security Architecture Guidelines Ref: Sebi Mutual Funds Regulations 2026 Acceptance: Processes 10,000 events/second with zero message loss and at-least-once delivery guarantees.
REQ-BE-04	Automated Regulatory Telemetry & FIU-IND Reporting Pipeline Implement automated transaction monitoring generating Cash Transaction Reports (CTR) and Suspicious Transaction Reports (STR) formatted for FIU-IND FINnet 2.0 batch exports.	Must Have	Regulation: Prevention of Money Laundering Act (PMLA 2002) & FIU-IND Reporting Directives Ref: 1556052828Rbi-Handbookofregulationsataglance-27.2.2025-3-76 Acceptance: Flags suspicious velocity or threshold anomalies (>₹10 Lakhs) within 60 seconds.
REQ-BE-05	External Banking Switch & Partner API Gateway Orchestrate external API integrations (NPCI, sponsor banks, credit bureaus, CKYCR) via mutual TLS (mTLS) with token-bucket rate limiting and circuit breaking.	Must Have	Regulation: RBI Guidelines on Outsourcing & Digital Lending Partner Oversight Acceptance: Restricts unauthorized partner API requests; enforces 150 req/min rate limit per tenant.
REQ-BE-06	Real-Time Automated Reconciliation Engine Deploy automated 2-way and 3-way reconciliation jobs matching application ledger entries against sponsor bank settlement files and clearing reports.	Must Have	Regulation: RBI Harmonisation of Turn Around Time (TAT) and Customer Compensation Acceptance: Identifies discrepancies and triggers automated compensatory reversals within T+1 day.

Pillar: Authentication, Identity & KYC Governance (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-AUTH-01	RBI Video-based Customer Identification Process (V-CIP) Execute automated customer identification with live facial liveness detection, Indian geo-coordinate validation, and real-time PAN verification.	Must Have	Regulation: RBI KYC Master Direction - V-CIP Amendment Ref: Fintech Regulatory Sandbox Pilot Problem Statement Acceptance: Rejects recorded spoof video attempts; verifies GPS coordinates within Indian territorial borders.
REQ-AUTH-02	UIDAI Aadhaar Data Vault (ADV) Hardware Security Isolation Store 12-digit Aadhaar numbers exclusively inside an isolated Hardware Security Module (HSM) Aadhaar Data Vault (ADV), substituting database references with tokens and masking first 8 digits.	Must Have	Regulation: UIDAI Aadhaar Data Vault Circular (2017) & Masking Directive Ref: Sebi Mutual Funds Regulations 2026 Acceptance: Zero plaintext 12-digit Aadhaar numbers persist in operational application tables or logs.
REQ-AUTH-03	Central KYC Registry (CKYCR) & NSDL PAN Verification Query CKYCR and NSDL databases in real time to auto-populate customer verification records with minimum 80% fuzzy name-match scoring.	Must Have	Regulation: PMLA Rules 2005 & CKYCR Operating Guidelines Ref: 1556052828Rbi-Handbookofregulationsataglance-27.2.2025-3-76 Acceptance: Returns active PAN status and CKYC ID within 2 seconds of query.

REQ-AUTH-04	Dynamic Two-Factor Authentication (AFA) with Device Binding Mandate Additional Factor of Authentication (AFA) binding device IMEI/MAC, registered mobile number, and transaction amount hash for all financial actions.	Must Have	Regulation: RBI Master Direction on Digital Payment Security Controls Sec 4 Acceptance: <i>Terminates active session upon hardware fingerprint mismatch; enforces 180s OTP expiration window.</i>
REQ-AUTH-05	Role-Based Access Control (RBAC) with Maker-Checker Controls Implement fine-grained RBAC with dual-authorization maker-checker workflows for high-value actions, administrative adjustments, and compliance overrides.	Must Have	Regulation: RBI IT Governance & Cyber Security Framework Acceptance: <i>Prohibits maker from approving own transactions; logs tamper-proof authorization signatures.</i>

Pillar: Information Security & Statutory Compliance (7 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-SEC-01	KYC Compliance Engine Automated KYC verification engine using Aadhaar UIDAI API for farmer applicants, generating digital identity certificates (DID) and storing them securely in a separate DID registry database	Must Have	Regulation: SEBI (Digital Ledgers and Registers Verification) Regulations, 2021 Acceptance: <i>KYC portal generates 1000 DID certificates for eligible farmers within 7 days, each certified by UIDAI with unique DID number and issuer ID</i>
REQ-SEC-02	Fraud Detection Module Machine learning model trained on historical DCI transaction data to detect fraudulent peer-to-peer transfers, flagging suspicious activity for manual review	Should Have	Regulation: RBI Guidelines on Digital Cash Instruments (DCI) - Operational Framework for Issuers Acceptance: <i>Fraud detection module detects and flags 90% of simulated fraudulent transactions out of 100 attempted transfers, reducing false positives through machine learning anomaly detection</i>
REQ-SEC-03	FIPS 140-2 Level 3 HSM Field-Level Encryption Encrypt all sensitive financial PII, account credentials, and transaction secrets at rest using AES-256-GCM managed by certified Hardware Security Modules (HSM).	Must Have	Regulation: RBI Digital Payment Security Controls Sec 5.2 Ref: Fintech Regulatory Sandbox Pilot Problem Statement Acceptance: <i>Database backups contain zero plaintext PII; cryptographic keys rotate automatically every 90 days.</i>
REQ-SEC-04	TLS 1.3 In-Transit Transport Security & SSL Pinning Enforce TLS 1.3 with forward-secrecy cipher suites across all endpoints, combined with dynamic certificate pinning on Android and iOS client applications.	Must Have	Regulation: NPCI Mobile Application Security Standards v2.4 Ref: Sebi Mutual Funds Regulations 2026 Acceptance: <i>Prevents MITM interception proxies from inspecting or modifying client-server payloads.</i>
REQ-SEC-05	Immutable WORM Forensic Audit Logging Maintain tamper-proof, append-only audit logs for all administrative actions, financial transactions, and authentication events with microsecond NTP time sync, preserved for 10 years.	Must Have	Regulation: CERT-In Cyber Security Direction No. 20(3)/2022-CERT-In & PMLA Sec 12 Ref: 1556052828Rbi -Handbookofregulationsatag glance-27.2.2025-3-76 Acceptance: <i>Logs stored in Write-Once-Read-Many (WORM) storage; prevents update or deletion by any system administrator.</i>
REQ-SEC-06	Web Application Firewall (WAF) & Rate Limiting Engine Protect ingress endpoints against OWASP Top 10 vulnerabilities, automated credential stuffing, and volumetric DDoS attacks via adaptive rate limiting.	Should Have	Regulation: OWASP ASVS Level 3 & RBI Cyber Security Guidelines Acceptance: <i>Blocks SQLi, XSS, and command injection at network perimeter; returns HTTP 429 upon rate limit breach.</i>
REQ-SEC-07	Card-on-File Tokenisation (CoFT) & Sensitive Token Lifecycle Implement token lifecycle management replacing sensitive financial identifiers with network tokens issued with explicit consent and de-tokenization controls.	Must Have	Regulation: RBI Tokenisation - Card Transactions Directive (RBI/2021-22/96) Acceptance: <i>Persistent storage contains solely encrypted tokens; raw card/PAN numbers discarded post-authorization.</i>

Pillar: Database Architecture & Data Residency (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-DB-01	100% Sovereign On-Soil Data Localization Host all primary databases, replicas, backups, and transaction logs exclusively within physical data centers located on Indian soil with zero foreign replication.	Must Have	Regulation: RBI Storage of Payment System Data Directive (RBI/2017-18/153) Ref: Fintech Regulatory Sandbox Pilot Problem Statement Acceptance: <i>Annual CERT-In empaneled System Audit Report (SAR) certifies 100% data residency inside Indian territory.</i>
REQ-DB-02	ACID Financial Ledger with Double-Entry Bookkeeping Implement relational tables with serializable transaction isolation enforcing double-entry ledger bookkeeping (Sum of Debits == Sum of Credits).	Must Have	Regulation: RBI Core Banking Accounting & Audit Norms Ref: Sebi Mutual Funds Regulations 2026 Acceptance: <i>Eliminates floating-point rounding errors; rolls back failed atomic transactions without orphan states.</i>
REQ-DB-03	DPDP Act 2023 Multilingual Consent Ledger & 72-Hour Data Erasure Record granular, unbundled consent timestamps and purposes in an immutable ledger, providing automated cascading data anonymization/erasure pipelines upon user revocation.	Must Have	Regulation: Digital Personal Data Protection Act, 2023 Sec 6 & Sec 12 Ref: 1556052828Rbi-Handbookofregulationsataglance-27.2.2025-3-76 Acceptance: <i>Executes non-statutory data erasure within statutory 72-hour window upon customer consent withdrawal.</i>
REQ-DB-04	Encrypted Archival & Statutory 10-Year Record Retention Orchestrate lifecycle policies archiving transactional and compliance records to cold encrypted storage with scheduled integrity verification hashes.	Must Have	Regulation: PMLA 2002 Sec 12 & Companies Act 2013 Acceptance: <i>Archives records older than 180 days with SHA-256 verification hash; guarantees 10-year retrieval readiness.</i>
REQ-DB-05	High-Availability Synchronous Multi-Zone Clustering (RPO < 1m, RTO < 15m) Deploy active-standby database clusters spanning multiple availability zones with automated failover guarantees achieving RPO < 1 minute and RTO < 15 minutes.	Should Have	Regulation: RBI Business Continuity Management Directive Acceptance: <i>Primary node failure triggers automated promotion of standby replica in <90 seconds with zero data loss.</i>

Pillar: Resilience, Scalability & Disaster Recovery (5 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-RES-01	Circuit Breaker & Graceful Degradation on Switch Gateways Incorporate Resilience4j circuit breakers across all external banking switches and microservices to prevent cascading thread pool exhaustion.	Should Have	Regulation: RBI IT Disaster Recovery & Resilience Guidelines Ref: Fintech Regulatory Sandbox Pilot Problem Statement Acceptance: <i>Switch error rate > 50% opens circuit within 5 seconds, returning cached fallback response without thread exhaustion.</i>
REQ-RES-02	Geo-Redundant Disaster Recovery (DR) Site in Alternate Seismic Zone Establish hot/warm secondary Disaster Recovery infrastructure located in a geographically distant Indian seismic zone with biannual live failover drills.	Must Have	Regulation: SEBI CSCRF BCP/DR Mandate & RBI Cyber Security Framework Ref: Sebi Mutual Funds Regulations 2026 Acceptance: <i>Production workload successfully cuts over to secondary DR site within 2 hours during scheduled drill.</i>
REQ-RES-03	Automated Failed Transaction Reversal & ■100/Day Compensation Dispatcher Deploy automated reconciliation daemon detecting failed or timed-out transactions and issuing reversals within T+1 day, calculating mandatory statutory delay compensation.	Must Have	Regulation: RBI Harmonisation of TAT & Customer Compensation (2019) Ref: 1556052828Rbi-Handbookofregulationsataglance-27.2.2025-3-76 Acceptance: <i>Issues customer refund within 24 hours of failure; automatically calculates ■100/day compensation for delays.</i>

REQ-RES-04	Real-Time Velocity & Anomaly Fraud Detection Engine Deploy rule-based and ML scoring pipelines evaluating transaction frequency, device changes, and geolocation anomalies to flag suspicious transactions prior to settlement.	Could Have	Regulation: RBI Master Direction on Fraud Risk Management Acceptance: <i>Flags high-velocity anomaly within 500ms, triggering mandatory step-up authentication or analyst alert.</i>
REQ-RES-05	Automated Stress Testing & NPCI Certification Test-Bed Harness Embed continuous performance testing pipelines simulating peak Indian festival transaction volumes (minimum 5,000 TPS) with zero message drops.	Should Have	Regulation: NPCI Operational Preparedness Guidelines & SEBI Capacity Norms Acceptance: <i>Sustains 5,000 TPS benchmark with 99th percentile response latency under 300ms.</i>

Pillar: Feature Requirement (2 Specifications)

Req ID	Title & Detailed Specification	Priority	Statutory Citation & Acceptance Criteria
REQ-EXT-34	Offline Payment Capability Capability to make offline cash deposits into CBDC wallets, allowing verified farmers to receive produce payments without internet connectivity	Should Have	Regulation: RBI Guidelines on Digital Cash Instruments (DCI) - Operational Framework for Issuers Acceptance: <i>Offline payment capability successfully accepts 1000 cash deposits from verified farmers, each credited to their CBDC wallet within 24 hours</i>
REQ-EXT-35	Wallet Balance Management User interface for managing CBDC wallet balances, enabling verified farmers to deposit cash, make peer-to-peer payments, and view account balance	Should Have	Regulation: RBI Guidelines on Digital Cash Instruments (DCI) - Operational Framework for Issuers Acceptance: <i>Wallet management UI enables 1000 verified farmers to perform 1000 peer-to-peer transactions, each with a maximum value of ₹500, resulting in accurate balance updates</i>

3. Regulatory & Statutory Grounding (Indian Banking Frameworks)

Framework	Enforcement Level	Statutory Directive / Citation
RBI Digital Payment Security Controls (2021)	Core Banking & Payment Security	RBI Master Direction on Digital Payment Security Controls
NPCI UPI Operating Guidelines & AutoPay Rules	Clearing & Mandate Orchestration	NPCI Procedural Guidelines on UPI Mandates
RBI Sovereign Data Localization Directive (2018)	100% Indian Data Center Mandate	RBI/2017-18/153 Storage of Payment System Data
Digital Personal Data Protection Act, 2023	Customer Consent & Data Governance	DPDP Act 2023, Sections 6 & 8

4. Mandatory Security & Architectural Controls

Control ID	Title & Architecture Specification	Automated Verification Method
CTRL-SEC-01	Retail Cryptographic Transport & Rest Encryption Enforce TLS 1.3 for all in-transit communications and AES-256-GCM for sensitive fields.	Automated security scanning and cryptographic key rotation verification.
CTRL-AUTH-02	Dual-Factor Authentication & Device Fingerprint Binding Mandatory 2FA with device hardware fingerprint binding and session timeout enforcement.	Penetration testing verifying zero session hijack or credential bypass.
CTRL-AUD-03	Statutory Immutable Forensic Audit Logging Maintain append-only WORM audit logs with microsecond NTP timestamps for statutory retention period.	Cryptographic checksum validation & CERT-In audit verification.
CTRL-RES-04	Automated Circuit Breaker & High Availability Failover Deploy Resilience4j circuit breakers and active-standby database failover with RPO < 1m.	Live chaos drill and automated failover switchover test.

5. Traceability Matrix & Recommended SDLC Sprints

Req IDs	Target Sub-Module	SDLC Phase / Gate	Verification Deliverable
REQ-FE-01 - REQ-FE-05	P2P CBDC Offline Wallet Platform — Frontend & Mobile UI/UX	Sprint 1 / Frontend Milestone	Frontend & Mobile UI/UX Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-BE-01 - REQ-BE-06	P2P CBDC Offline Wallet Platform — Backend Architecture & Core APIs	Sprint 2 / Backend Milestone	Backend Architecture & Core APIs Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-AUTH-01 - REQ-AUTH-05	P2P CBDC Offline Wallet Platform — Authentication, Identity & KYC Governance	Sprint 3 / Authentication, Milestone	Authentication, Identity & KYC Governance Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-SEC-01 - REQ-SEC-07	P2P CBDC Offline Wallet Platform — Information Security & Statutory Compliance	Sprint 4 / Information Milestone	Information Security & Statutory Compliance Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-DB-01 - REQ-DB-05	P2P CBDC Offline Wallet Platform — Database Architecture & Data Residency	Sprint 5 / Database Milestone	Database Architecture & Data Residency Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
REQ-RES-01 - REQ-RES-05	P2P CBDC Offline Wallet Platform — Resilience, Scalability & Disaster Recovery	Sprint 6 / Resilience, Milestone	Resilience, Scalability & Disaster Recovery Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>

REQ-EXT-34 - REQ-EXT-35	P2P CBDC Offline Wallet Platform — Feature Requirement	Sprint 7 / Feature Milestone	Feature Requirement Architecture & Verification Suite <i>Automated Integration Test & Security Signoff</i>
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